Benjamin Fankhauser- Engineering Portfolio

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South Sound Racing
UW Tacoma FSAE

Chassis Lead | Jun 2026 - Present | Preliminary design complete

Joined UW Tacoma's newly formed Formula SAE team prior to transferring to UW Seattle from Green River. Subsequently promoted to Chassis Lead, responsible for preliminary chassis design and integration.

Chassis Development

- Translated the 2027 FSAE rulebook and the 2027 structural equivalency spreadsheet's structural requirements into a baseline spaceframe chassis.
- Benchmarked other teams' chassis layouts to inform geometry decisions.
- Classified the rear structure separately at Size C (the structural tubing minimum). Under F.3.3, tubing that doesn't meet structural size can be smaller, so it can be downsized later if needed.

Team Initiative

Sourced team access to an 800 x 800 x 1000 mm 3D printer through an external personal connection. Established the print request process and coordinated fabrication for the team.

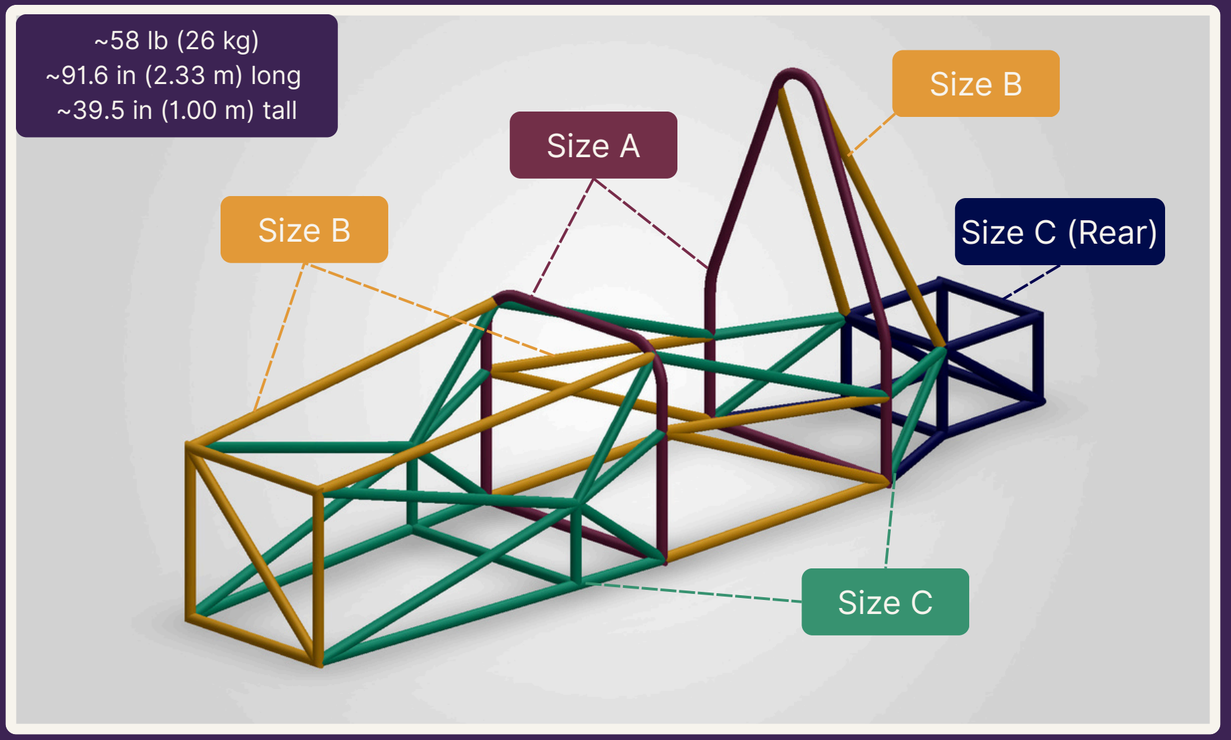
Subsystem Integration

Provided early drivetrain and cooling/intake packaging input during initial vehicle layout. Now coordinating with subteam leads to integrate mounting and packaging requirements into the chassis.

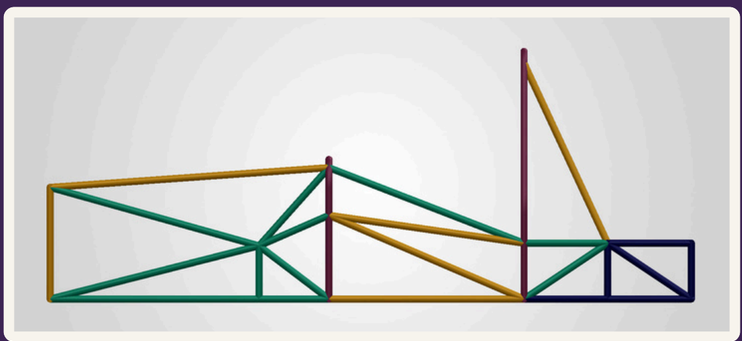
Current Development

- Revising the model to integrate mounting and packaging requirements from suspension and other subteam leads.
- Next: FEA (torsional stiffness and load cases) and a mock build.

Last Updated: 11/07/2026



Size A	1 in x 0.095 in	Front/main hoops
Size B	1 in x 0.065 in	Front bulkhead, side impact, bracing
Size C	1 in x 0.049 in	Bulkhead supports, bracing supports
Size C (Rear)	1 in x 0.049 in	Rear structure drawn Size C, may be downsized

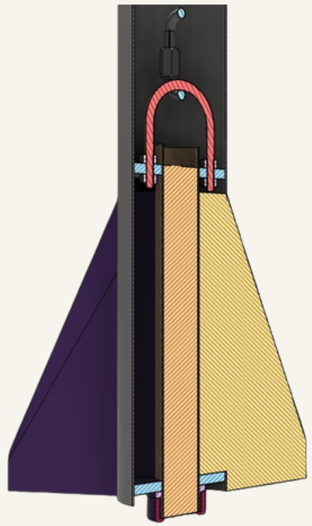


Design Foundation:

- Colors show tube size required by the 2027 FSAE Rules (F.3.2.1, F.3.3, and F.3.4.1)
- Mass, length, and height measured from bare tubes only (no welds, brackets, or panels)
- Current Material: AISI 4130 Steel

L1 High-Power Certification Rocket

Sole Designer and Fabricator
Jul 2026 - Present | Build in progress



Overview & Methods

- **Project Description:** End-to-end design and fabrication of a **47" long, 3.1" OD motor ejection high-power rocket** for Tripoli L1 certification, begun with no prior experience. Certification flight pending.
- **Tools & Methods Utilized:** OpenRocket, custom CAD, FDM 3D printing (PETG), laser cutting, precision fit-up, structural bonding.

Key Design Decisions

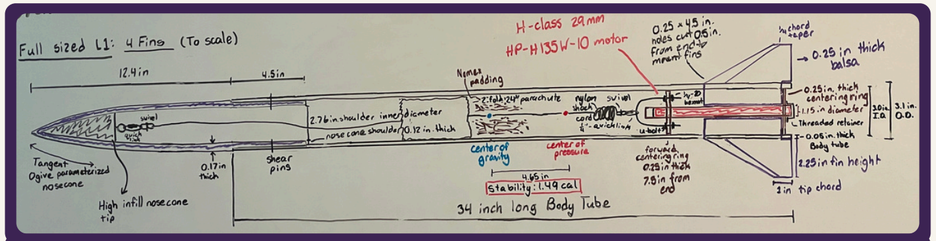
- **Challenges Faced:** Adding actual recovery hardware masses dropped simulated stability to under **0.9 cal**, below the safety threshold.
- **Results and Outcomes:** Redesigning the fins (4 smaller to 3 larger) with larger tip chord/span restored stability to **1.5+ cal**, but revealed an unsafe flutter margin **(0.86x)** at the original thickness (**1/4"**). Switching to **3/16"** birch plywood fixed flutter **(1.45x)** with negligible stability impact and no carbon/fiberglass layer, which is being saved for my L2 design. CAD and slicer-based nose cone weight estimates added mass and brought stability up to a final simulated **1.99 cal.**
- **Aerodynamic Profile Tooling:** Designed and 3D-printed a custom sanding fixture to transform flat laser-cut fin stock into a 1/4-chord tapered airfoil profile, matching geometry used in the flutter analysis.



Subscale Validation Build

OpenRocket Simulation Data

Apogee	787 m
Max velocity	157 m/s
Time to apogee	11.9 s



Preliminary Design Pre Fin Change

Red Ryder Revival
Washington Aerospace Club

Manufacturing & Propellant Team | May 2026 - Present | Flown Sep 28

Overview

Contributing member on Red Ryder, a 22 ft, 385 lb high-power rocket built by the Washington Aerospace Club. Joined after the design phase and contributed hands-on manufacturing and propellant work.

Flight Outcome: Flew September 28 at BALLS 34 in Black Rock Desert, Nevada, to about 4,500 ft on a separate, lower-power motor because the remixed propellant wasn't authorized in time. The custom propellant is held for a future flight.



Contributions

- **AV Bay Assembly:** Cut out and assembled bulkheads, with threaded inserts to attach through-rods holding the AV slide rails.
- **General Fabrication:** Contributed across multiple build sessions, with epoxy work and general manufacturing based on drawings and specs.
- **Propellant Remix:** Worked with the WAC propulsion division (WARP) to remix the team's solid propellant grain under the supervision of a professional solid propulsion engineer to the original design recipe after an earlier batch was mixed incorrectly and failed static testing. Followed strict protocols to minimize friction and ignition risks.



Active-Ballast Submarine System

Two-person team | Jun 2026 - Present | Prototype tested, hull redesign in planning

Overview

Began as a Green River Engineering Club ROV project, continued with a partner into a true submarine. Designed the stepper-driven ballast mechanism and all mechanical components, then sourced, manufactured, and assembled the system, wiring included. Partner designed the control circuit and wrote the code.

Key Design Decisions

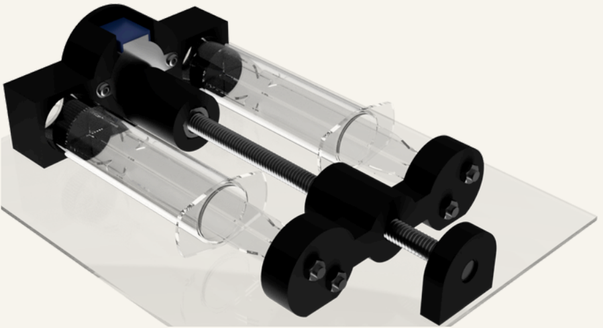
- **Mechanism Pivot:** Water pressure could back-drive the initial rack and pinion system when unpowered, so it was replaced with a threaded-rod leadscrew that resisted back-drive and drove two syringes at once without additional gearing, allowing even ballast distribution.

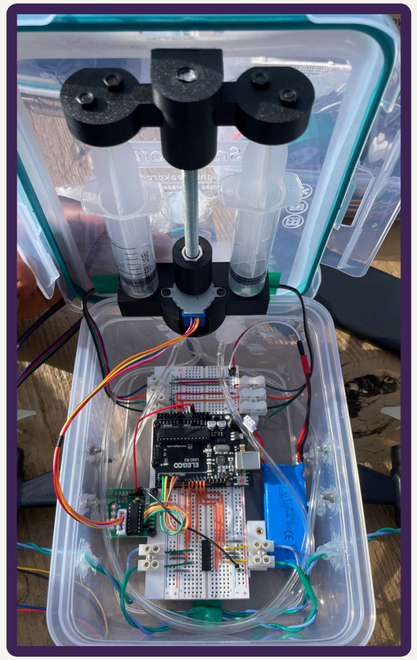
- **Motor Speed Fix:**
  - A coarser thread pitch cut full ballast travel from roughly 3-4 minutes to ~2 minutes.

- **Manufacturing Tolerance:**
  - Compensated for 0.15-0.2 mm of print undersizing to maintain tight component fit.

Test Results and Next Steps

- **Result:** The ballast mechanism worked with no leaks in the full-system water test.
- **Buoyancy:** Neutral buoyancy required all of our steel pellets (**1,050 g**) plus additional weight, substantially exceeding initial estimates.
- **Weight Distribution:** Leadscrew interference forced the tether off center, causing the weight to shift as the system was lowered.
- **Next Steps:** Develop a purpose-built, 3D-printed, epoxy-sealed hull with optimized ballast placement and a centered tether attachment.





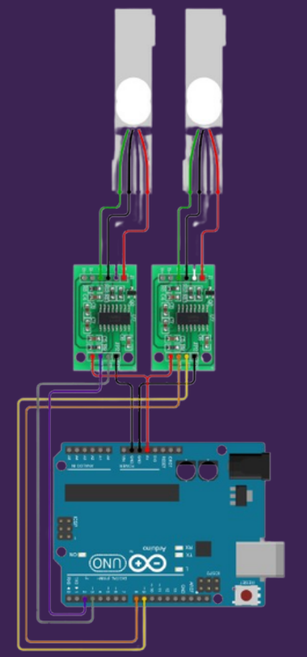
Wind Tunnel Instrumentation

Independent Project | Planned 2027
Concept and interface circuit complete

Designed a dual-channel load-cell concept to independently measure aerodynamic drag and downforce on scale-model wind tunnel tests.

Concept: One load-cell per axis, each read through its own HX711 amplifier by an Arduino, allowing drag and downforce readings to be measured separately.

Status: Load cells and amplifiers sourced, sensor interface schematic designed. Bench prototype next.



Trebuchet

Independent Project, Open-Ended Physics Coursework | Nov 2025 | Complete

Designed and fabricated from scratch to test how counterweight mass affects launch range, later modeled in Fusion 360 for practice.

Data Collection: Recorded launches against a measured grid backdrop; calculated distance per frame, then converted to velocity using frame rate.

Result: Across 4 counterweight masses **(501-726 g)**, average range increased from **210** to **337 cm.** Measured range consistently ran **6-17%** below kinematic prediction, likely due to drag and friction.

